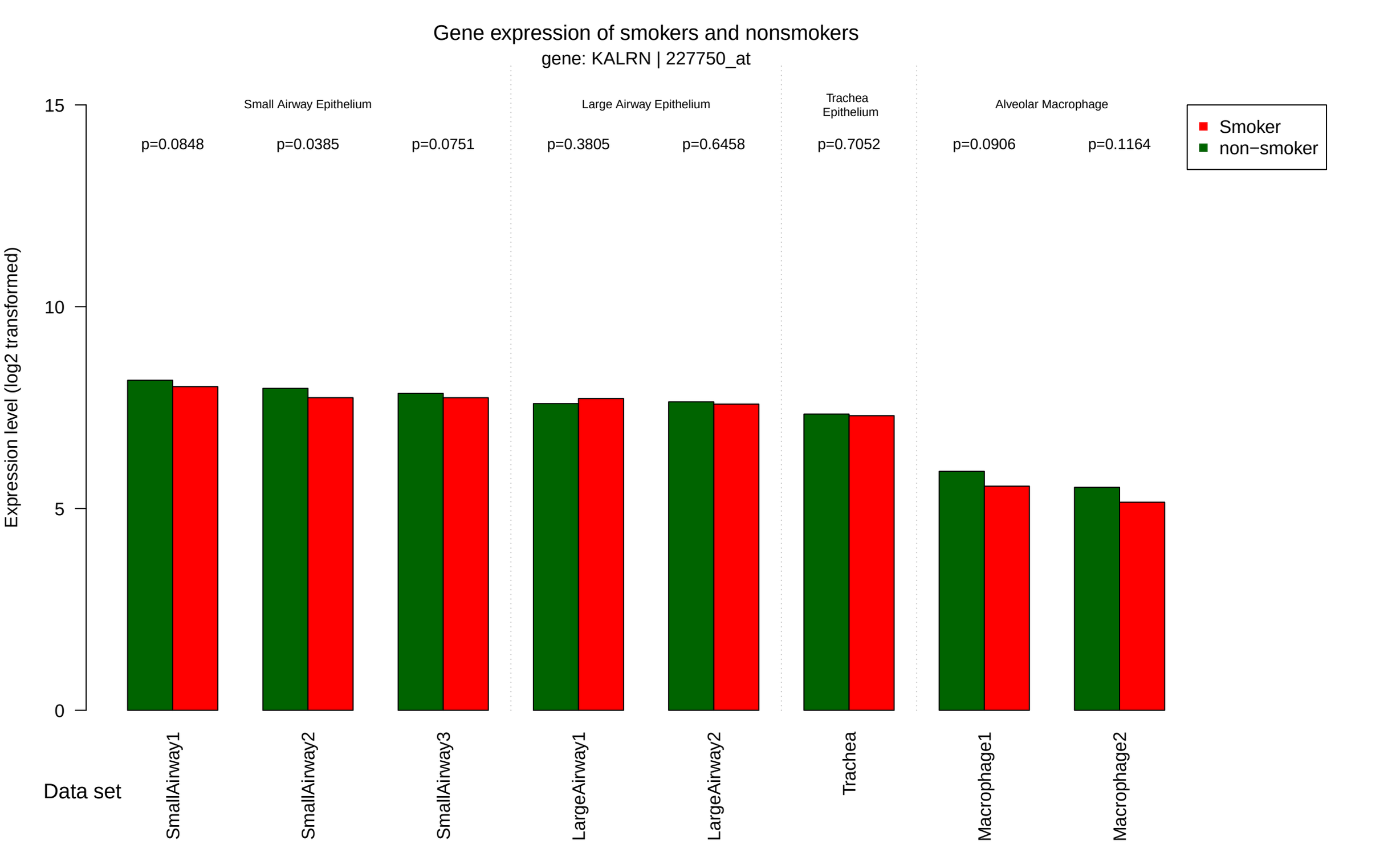
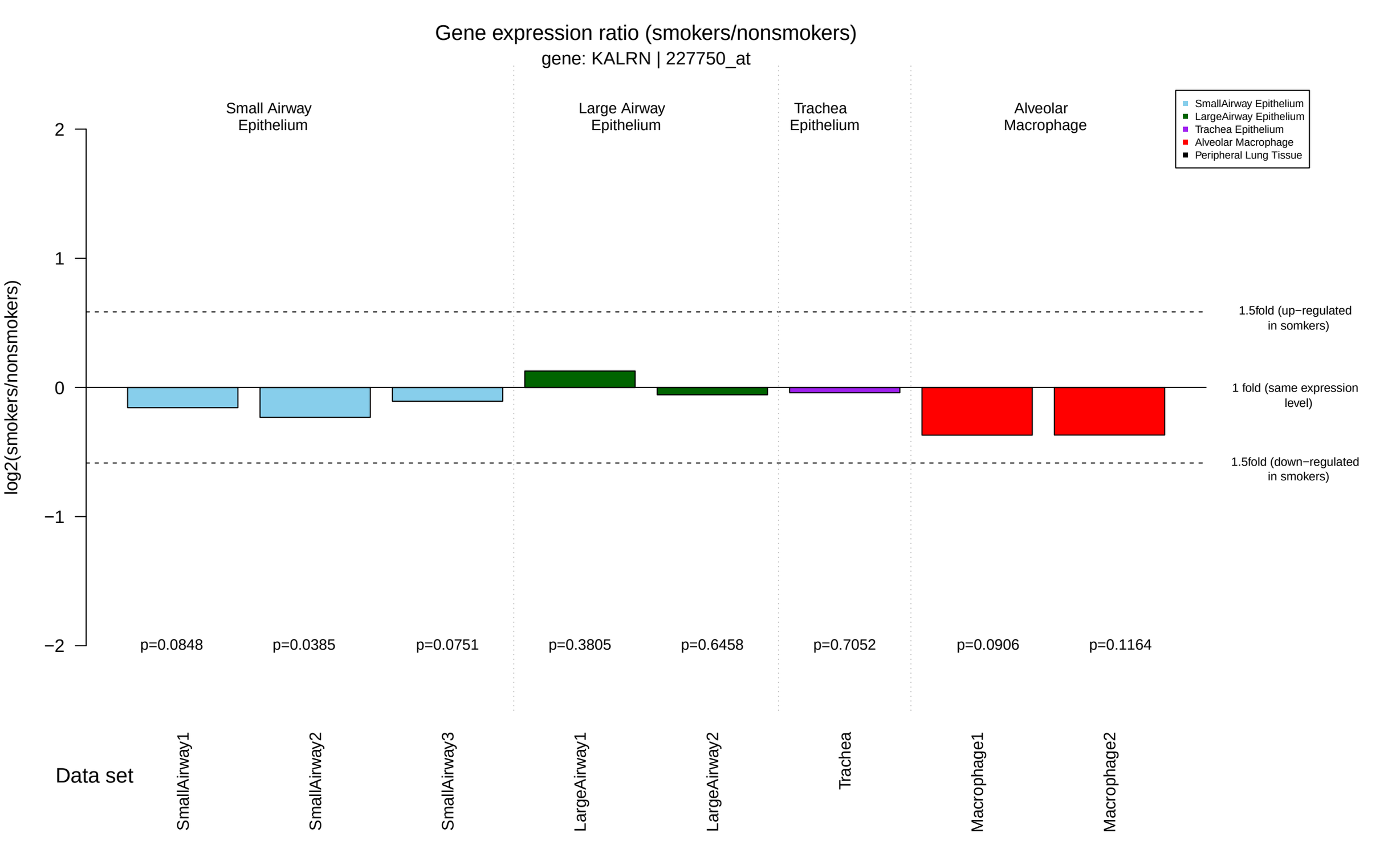
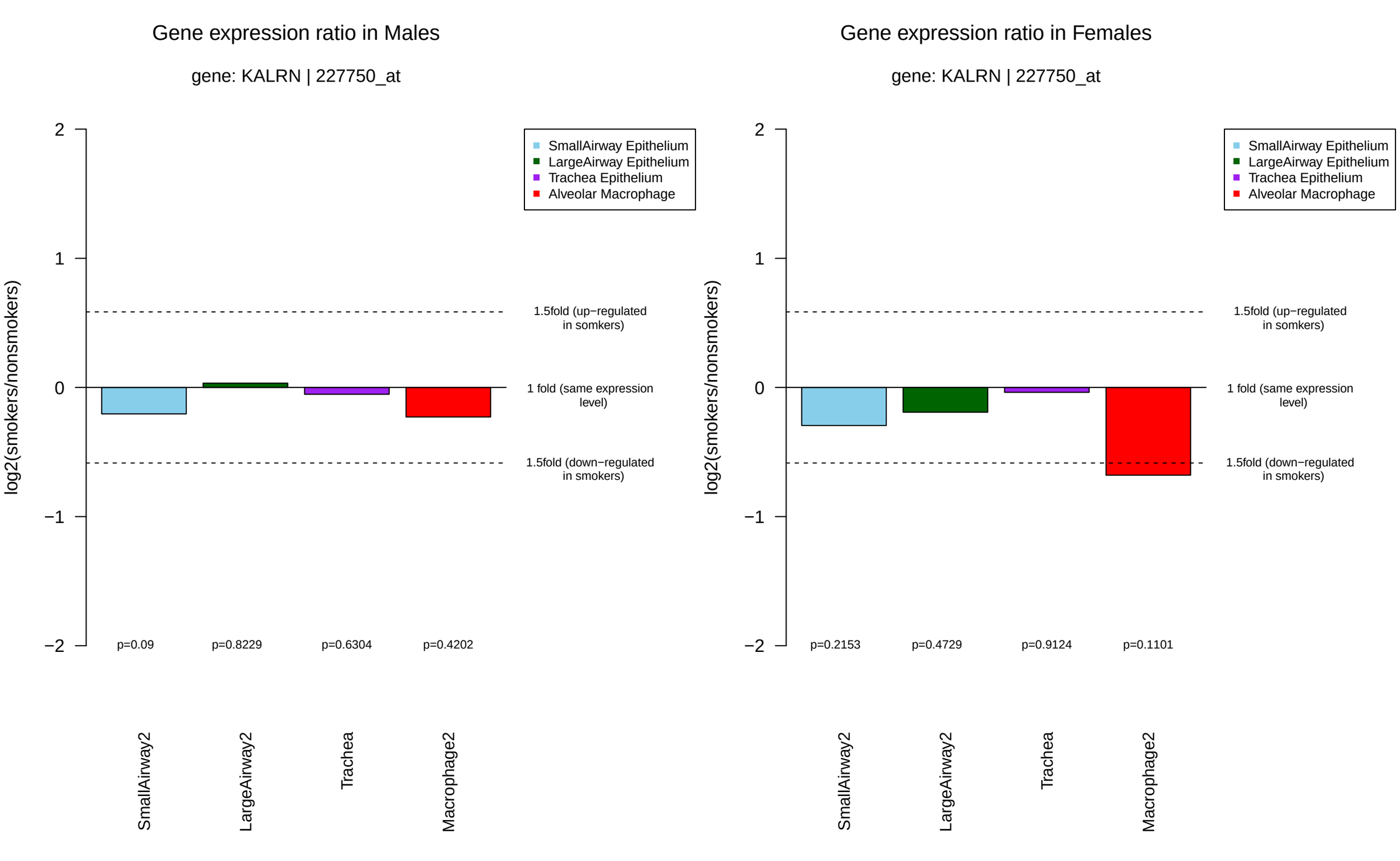
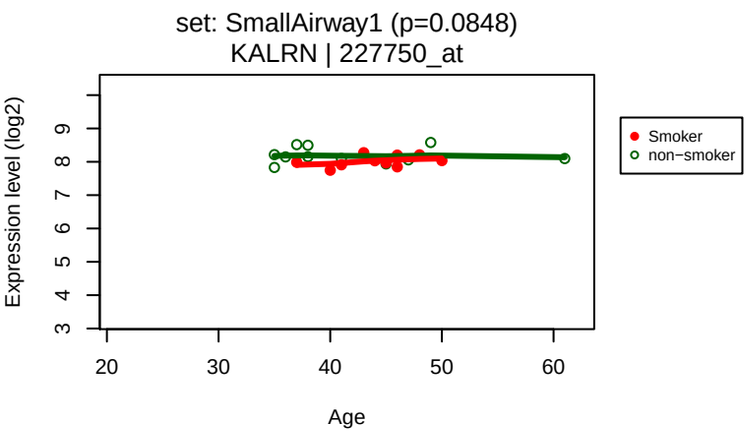
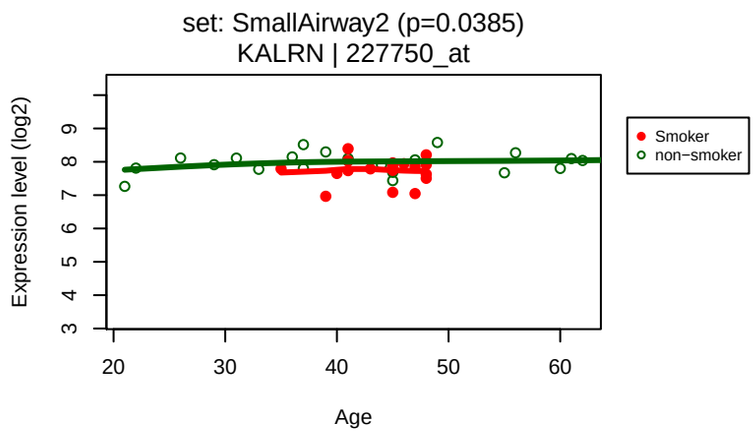
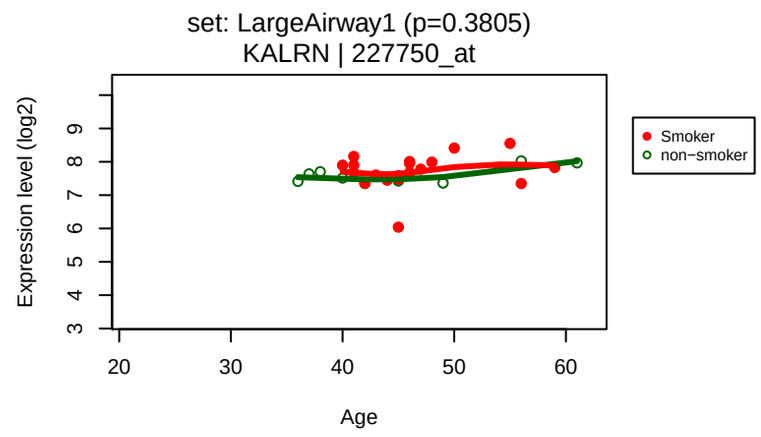

Gene expression of smokers and nonsmokers
gene: KALRN | 227750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0848
p=0.0385
p=0.0751
p=0.3805
p=0.6458
p=0.7052
p=0.0906
p=0.1164
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

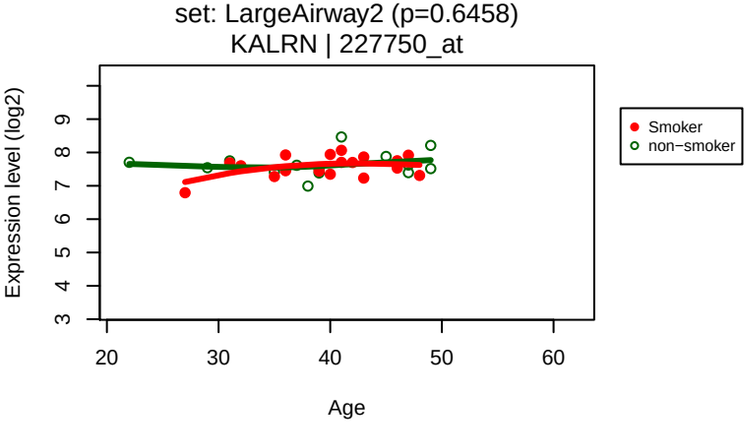

Gene expression ratio (smokers/nonsmokers)
gene: KALRN | 227750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0848
p=0.0385
p=0.0751
p=0.3805
p=0.6458
p=0.7052
p=0.0906
p=0.1164
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

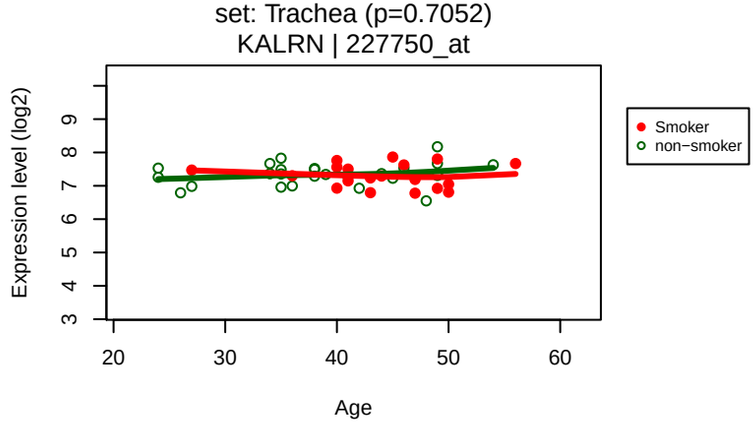

Gene expression ratio in Males
gene: KALRN | 227750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.09
p=0.8229
p=0.6304
p=0.4202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KALRN | 227750_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2153
p=0.4729
p=0.9124
p=0.1101
SmallAirway2
LargeAirway2
Trachea
Macrophage2

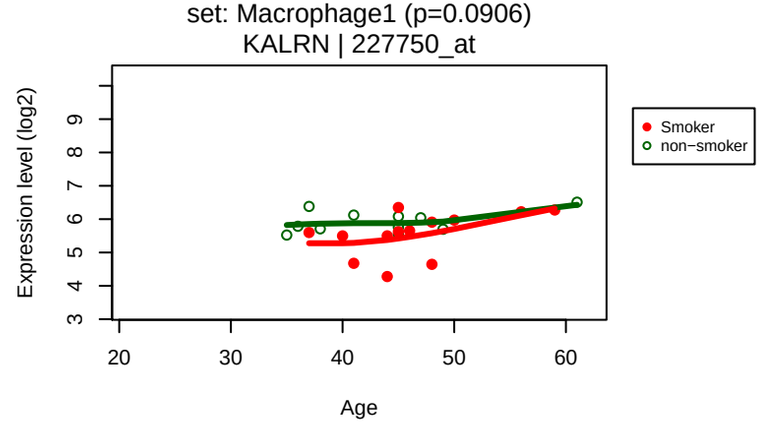

set: SmallAirway1 (p=0.0848)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker

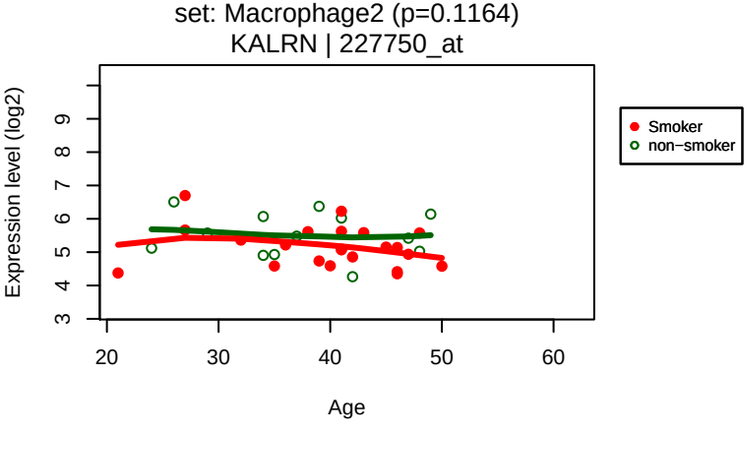

set: SmallAirway2 (p=0.0385)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker

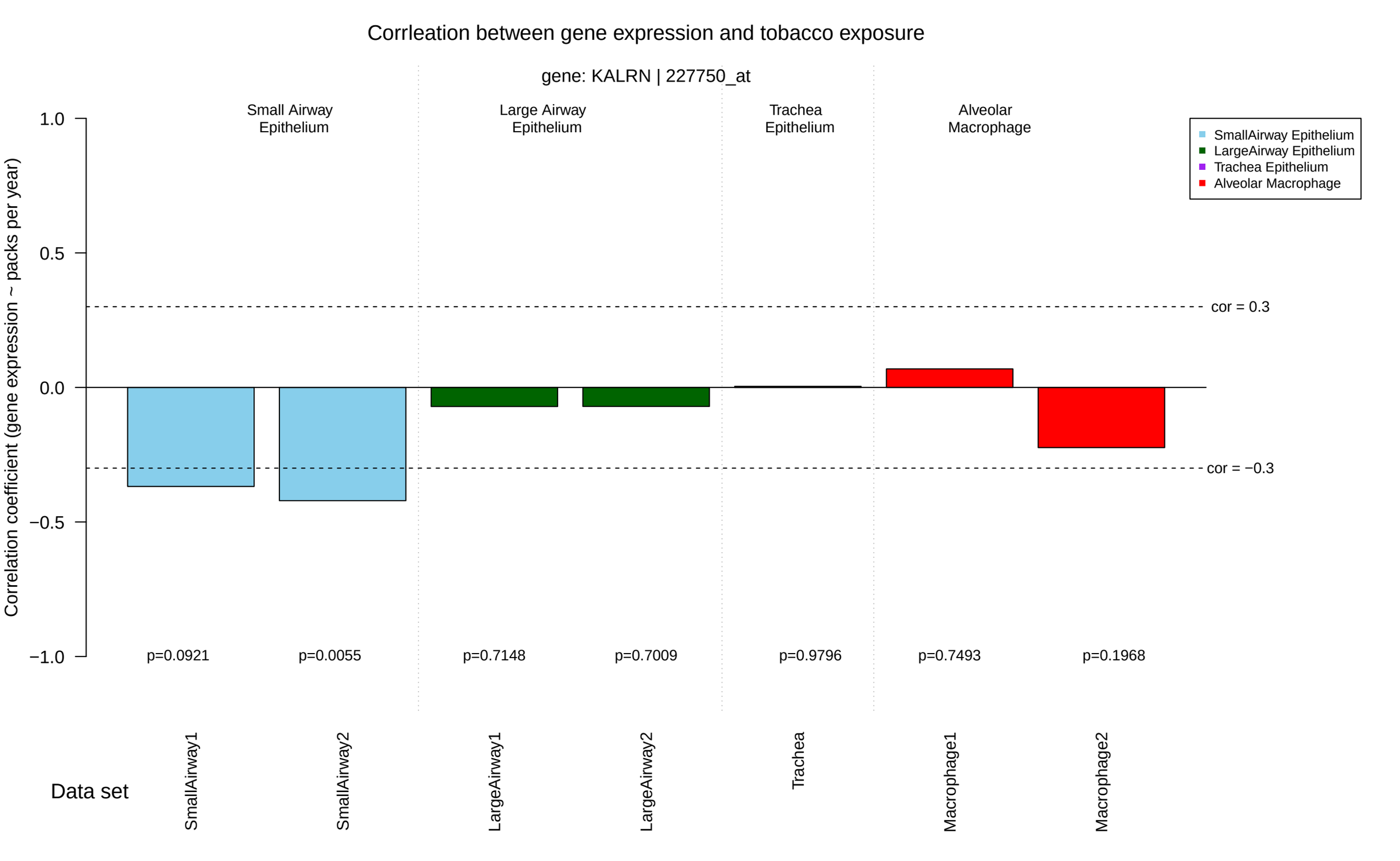

set: LargeAirway1 (p=0.3805)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6458)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7052)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0906)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1164)
KALRN | 227750_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KALRN | 227750_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0921
p=0.0055
p=0.7148
p=0.7009
p=0.9796
p=0.7493
p=0.1968
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2